SafeSignal: Detecting Hidden Drug Side-Effects from Patient Reviews using NLP

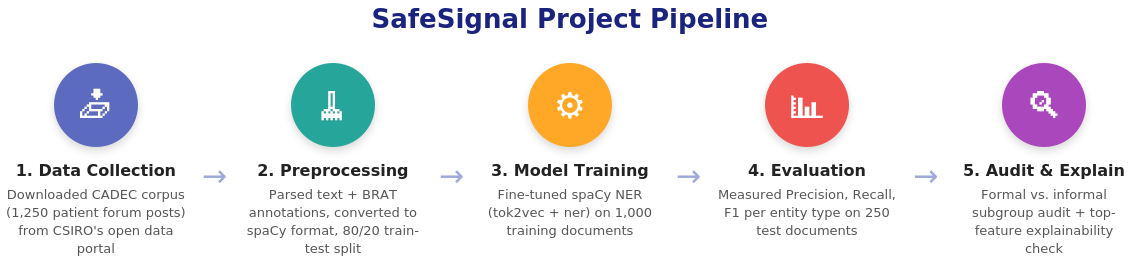
Project Write-up (Build Component)

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1. Problem Statement

Clinical trials cannot capture all adverse drug reactions (ADRs) before a drug reaches the market. Patients often report side-effects informally on health forums, but this valuable data goes largely unanalyzed. This project builds an NLP pipeline to automatically extract Drug, Adverse Reaction (ADR), Disease, Symptom, and Finding entities from patient-written text, enabling scalable detection of hidden ADR signals.

Project Pipeline Overview



2. Dataset

- Source: CADEC (CSIRO Adverse Drug Event Corpus)
- Size: 1,250 forum posts from AskaPatient.com, covering 12 drugs (e.g., Lipitor, Voltaren, Arthrotec)
- Annotations: BRAT-format entity spans across 5 categories - Drug, ADR, Disease, Symptom, Finding
- License: Publicly available via CSIRO's open research data portal (data.csiro.au)

3. Preprocessing

- Parsed raw .txt posts and corresponding .ann annotation files
- Converted character-offset entity spans into spaCy-compatible token spans using char_span()
- Filtered overlapping entity spans using spaCy's filter_spans()
- Split data into 1,000 training documents and 250 test documents (80/20 split, random seed = 42)
- Approximately 4% of entity annotations were skipped due to tokenization misalignment - a known challenge with informal, punctuation-inconsistent forum text

4. Approach / Model

- Framework: spaCy (blank English pipeline with tok2vec + ner components)
- Training: Fine-tuned from scratch on CADEC training data using spaCy's CLI training pipeline, optimized for efficiency (CPU-based)
- Evaluation metric: Precision, Recall, F1 (entity-level, per class and overall)

5. Results

Overall test set performance:

Entity	Precision	Recall	F1
Drug	0.94	0.88	0.91
ADR	0.65	0.65	0.65
Finding	0.43	0.29	0.35
Disease	0.33	0.12	0.18
Symptom	0.50	0.08	0.14

The model performs strongly on Drug and ADR entities - the two most clinically important categories for this task - while showing weaker performance on rarer entity types (Disease, Symptom), likely due to class imbalance in the training data. Full audit findings (formal vs. informal subgroup analysis and explainability) are detailed separately in the model card.

5.1 Sample Prediction

To verify the model works correctly on real text, it was tested on a new, unseen sentence:

"I took Lipitor and started having severe muscle pain and memory problems."

- Lipitor -> Drug
- severe muscle pain -> ADR
- memory problems -> ADR

This confirms the model correctly distinguishes the drug name from the reactions it caused, even in a casually phrased sentence.

5.2 Result Visualizations

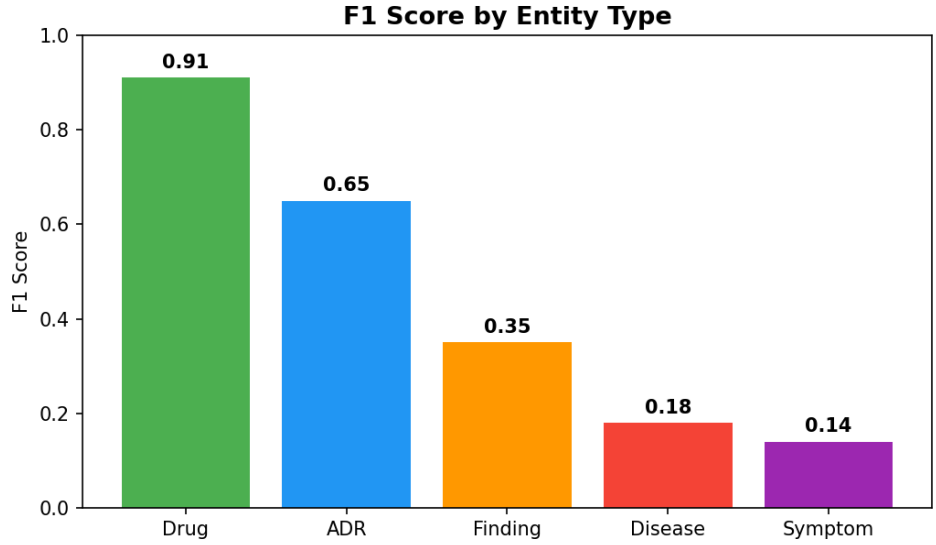


Figure 1: F1 score by entity type. Drug and ADR - the two clinically critical categories - are detected far more reliably than the rarer classes.

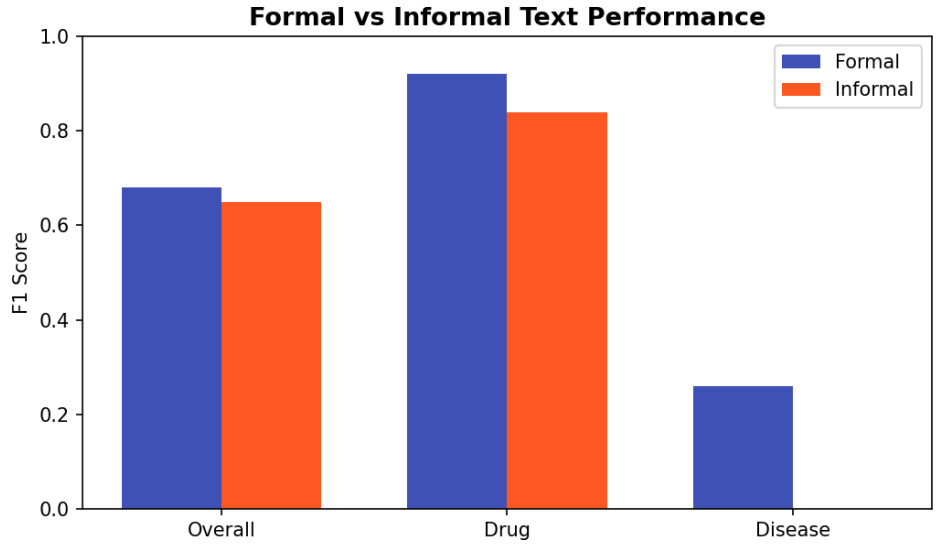


Figure 2: Formal vs. informal text performance. The model's accuracy drops on informal, casually written posts, most sharply for Disease detection (see model card for the full audit).